10. What is dispersion? Obtain the equation for dispersive power of a medium.

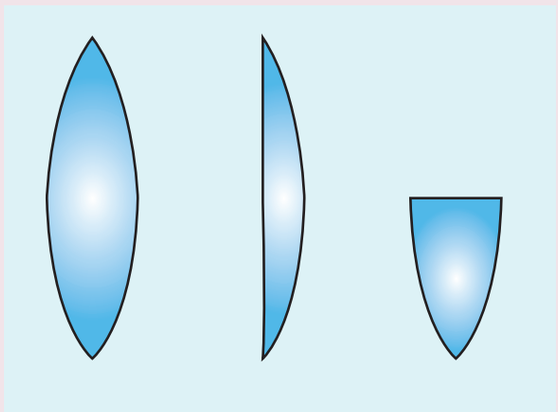
Conceptual Questions

1. Why are dish antennas curved?

2. What type of lens is formed by a bubble inside water?

3. Is it possible for two lenses to produce zero power?

4. A biconvex lens has focal length f and intensity of light I passing through it. What will be the focal length and intensity for portions of lenses obtained by cutting it vertically and horizontally as shown in figure?



5. Why is yellow light preferred to during fog?

Numerical Problems

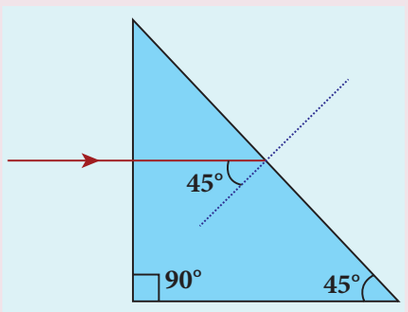
1. An object of 4 cm height is placed at 6 cm in front of a concave mirror of radius of curvature 24 cm. Find the position, height, magnification and nature of the image.

[Ans: $v = 12$ cm, $h' = 8$ cm, $m = 2$, image is erect, virtual, twice the height of object formed on right side of mirror.]

2. An object is placed in front of a concave mirror of focal length 20 cm. The image formed is three times the size of the object. Calculate two possible distances of the object from the mirror.

[Ans: with $+m$, $u = -40/3$ cm and with $-m$, $u = -80/3$ cm]

3. A beam of light consisting of red, green and blue is incident on a right-angled prism as shown in figure. The refractive index of the material of the prism for the above red, green and blue colours are 1.39, 1.44 and 1.47 respectively. What are the colours suffer total internal reflection?



[Ans: green and blue suffer total internal reflection]

4. An object is placed at a certain distance from a convex lens of focal length 20 cm. Find the object distance if the image obtained is magnified 4 times.

[Ans: $u = -15$ cm.]

5. Obtain the lens maker's formula for a lens of refractive index n_2 which is separating two media of refractive indices n_1 and n_3 on the left and right respectively.

$$\left[\text{Ans } \frac{n_3}{v} - \frac{n_1}{u} = \frac{(n_2 - n_1)}{R_1} + \frac{(n_3 - n_2)}{R_2}\right.$$

$$\left.\frac{1}{f} = \frac{1}{n_3}\left(\frac{(n_2 - n_1)}{R_1} + \frac{(n_3 - n_2)}{R_2}\right)\right]$$

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